

How to treat autism in children?

Evidence searched on 2026-10-08 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line The retrieved evidence supports **exercise interventions** (improving motor skills) and **behavioral therapies** (e.g., naturalistic developmental interventions) as effective approaches for children with autism spectrum disorder (ASD). A **modified ketogenic diet combined with rehabilitation** showed short-term reductions in behavioral and symptom severity, but this remains a targeted, investigational strategy.

Most important recent finding Exercise significantly improved overall, gross, and fine motor skills in children with ASD, with dose-response relationships identified for overall and gross motor skills [1].

Immediate context Behavioral interventions (e.g., caregiver-assisted group-based naturalistic developmental behavioral interventions) are promising for early developmental gains, while dietary approaches like the modified ketogenic diet show preliminary behavioral benefits but lack long-term evidence. No broad-spectrum treatment is established for core ASD symptoms.

Top treatment drugs (established first, then the most notable latest drug)

Drug	Indication named in the FDA label	Source
Aripiprazole	"Irritability Associated with Autistic Disorder"	DailyMed
Risperidone	"Treatment of irritability associated with autistic disorder"	DailyMed
NT1164 (latest, investigational)	Being tested in a registered Phase 3 trial, Not yet recruiting; not an approved treatment	NCT07257939

Latest findings

- **2026 Dec:** Observational study [2] – Altered sleep spindle characteristics were significantly associated with autism spectrum disorder in children, with reduced spindle amplitude, number, and density.
- **2026 Dec:** Observational study [3] – A 12-month case study reported sustained reductions in core ASD symptoms (e.g., ATEC scores decreased by 81.3%) following bilateral dual-target deep brain stimulation.
- **2026-09-22:** Case study [4] – Crohn's disease was misdiagnosed in an adolescent with ASD due to communication barriers, highlighting gastrointestinal symptom masking.
- **2026-09-18:** Meta-analysis [1] – Exercise interventions improved overall, gross, and fine motor skills in children with ASD, with dose-response effects identified.

Current evidence

Approach	Evidence level	What the evidence shows	Reference
Exercise interventions	Evidence-supported but limited	Exercise significantly improved motor skills (overall, gross, fine) in children with ASD, with dose-response effects.	[1]
Naturalistic Developmental Behavioral Interventions (NDBIs)	Evidence-supported but limited	Caregiver-assisted group-based NDBIs showed promising early developmental improvements, though nonverbal gains were modest.	[5]

Approach	Evidence level	What the evidence shows	Reference
Modified ketogenic diet + rehabilitation	Emerging	A 2-month modified ketogenic diet combined with rehabilitation reduced ASD symptom severity (ABC/CARS scores) more than rehabilitation alone.	[6]
Extended reality (VR/AR) interventions	Emerging	VR/AR systems (e.g., training, therapy) showed growing research interest post-2020, with hybrid systems most common.	[7]
Deep brain stimulation (DBS)	Experimental or preliminary	A 12-month case study reported sustained reductions in core ASD symptoms (e.g., ATEC scores down 81.3%) post-DBS.	[3]
Intranasal oxytocin	Indirect evidence (children/adolescents)	No significant change in primary ABC-mSW scores vs. placebo in a Phase II trial (2021).	[8]
Sensory integration therapy	Insufficient evidence	No significant improvement in daily functioning vs. usual care in a 2022 RCT (n=106); carers reported subjective improvements.	[9]

Treatment Drug Landscape

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Doses appear only where an FDA label or source gives them for the use shown; each dose entry is a short summary of the label's dosing section for this condition, with every number checked against the label text (full text in the DailyMed link); individual dosing is a prescriber's decision.

Drug	Category	Indication	Dose / route (if sourced)	Approval status / date	Key evidence
Aripiprazole	Approved (FDA-labelled for this condition)	"Aripiprazole is indicated for the treatment of: • Schizophrenia • Irritability Associated with Autistic Disorder • Treatment of Tourette's Disorder"	Route: Oral. Pediatric Patients (6 to 17 years) The recommended dosage range for the treatment of pediatric patients with irritability associated with autistic disorder is 5 to 15 mg/day. Dosing should be initiated at 2 mg/day. The dose should be increased to 5 mg/day, with subsequent increases to 10 or 15 mg/day if needed. Dose adjustments of up to 5 mg/day should occur gradually, at intervals of no less than one week.	FDA-approved for the use shown; label effective 2026-03-05; original approval date not retrieved	FDA label. DailyMed

Drug	Category	Indication	Dose / route (if sourced)	Approval status / date	Key evidence
Risperidone	Approved (FDA-labelled for this condition)	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years)."	Route: Oral. Start with 0.25 mg per day for children weighing less than 20 kg, and 0.5 mg per day for those weighing 20 kg or more. After at least four days, increase to 0.5 mg per day for children under 20 kg or 1.0 mg per day for those weighing 20 kg or more, and maintain this dose for no less than 14 days. The dose may then be raised every 2 weeks or more in steps of 0.25 mg per day for children under 20 kg or 0.5 mg per day for those weighing 20 kg or more, with a maximum effective range of 0.5 mg to 3 mg per day. This dosing applies to children and adolescents weighing no less than 15 kg.	FDA-approved for the use shown; label effective 2026-06-25; original approval date not retrieved	FDA label: Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years). DailyMed

Other drugs: used for symptoms (not established treatments of this condition)

Off-label: a retrieved review or page describes use for specific symptoms or co-occurring conditions; the drug has no FDA-labelled indication for this condition. Drugs still being tested in trials are listed in the Clinical trials section below.

Drug	Indication	Key evidence
Melatonin	sleep problems in ASD	"Melatonin is commonly used to address sleep problems in ASD and appears to be effective." Pharmacological treatments in autism spectrum disorder: a narrative review / Journal of Psychopathology

Recently approved or updated (FDA, EMA and CDSCO announcements)

Date	Type	Announcement	What the regulator's page says
date not stated	Regulatory announcement	FDA Takes Action to Make a Treatment Available for Autism Symptoms / FDA	"The FDA is collaborating with GSK to broaden the existing Wellcovorin label," said George Tidmarsh, M.D., Ph.D., Director of the FDA's Center for Drug Evaluation and Research.
date not stated	Regulatory announcement	FDA Responds to Evidence of Possible Association Between Autism and Acetaminophen Use During Pregnancy / FDA	Food and Drug Administration today initiated the process for a label change for acetaminophen (Tylenol and similar products) to reflect evidence suggesting that the use of acetaminophen by pregnant women may be associated with an increased risk of neurological conditions such as autism and ADHD in children. "The FDA is taking action to make parents and doctors aware of a considerable body of evidence about potential risks associated with acetaminophen," said FDA Commissioner Marty Makary, M.D., M.P.H.

Page wording from the regulator's own site; read the page for the full announcement.

Safety warnings (FDA boxed warnings of the drugs listed above)

Drug(s) (FDA label)	Boxed warning, in the label's words
Lumateperone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS; and SUICIDAL THOUGHTS AND BEHAVIORS Increased Mortality in Elderly Patients with Dementia-Related Psychosis Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. CAPLYTA is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."
Sertraline	"Suicidality and Antidepressant Drugs Antidepressants increased the risk compared to placebo of suicidal thinking and behavior (suicidality) in children, adolescents, and young adults in short-term studies of major depressive disorder (MDD) and other psychiatric disorders. Anyone considering the use of Sertraline hydrochloride or any other antidepressant in a child, adolescent, or young adult must balance this risk with the clinical need."
Aripiprazole	"S: INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS AND SUICIDAL THOUGHTS AND BEHAVIOURS WITH ANTIDEPRESSANT DRUGS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Aripiprazole is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."
Risperidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Risperidone tablets are not approved for the treatment of patients with dementia-related psychosis."

These are the labels' own boxed warnings, not a complete safety review: read the full label.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT07863752	Stronger Together Mindfulness-Based Intervention; Psychoeducation Program	Phase not applicable	Completed (2026-10-07)	Completed; results not posted/retrieved.
NCT06234501	Pyridoxal 5'-Phosphate (100 mg); Placebo	Phase not applicable	Recruiting (2026-10-07)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07076264	AquOTic	Phase not applicable	Recruiting (2026-10-07)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06659328	Structured Walking Intervention for Children with Autism Spectrum Disorder.	Phase not applicable	Completed (2026-10-06)	Completed; results not posted/retrieved.
NCT04903353	Comparison of Risperidone and Aripiprazole	Phase 4	Completed (2026-10-05)	Completed; results not posted/retrieved.
NCT07257939	NTI164; Placebo	Phase 3	Not yet recruiting (2026-09-30)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07285798	KarXT; KarX-EC	Phase 3	Not yet recruiting (2026-09-28)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07243561	hUC-MSC-Exos Nasal Spray	Phase 1/2	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT06229210	Lumateperone	Phase 3	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT03279471	Sertraline; CBT/BIACA	Phase 2	Completed (2026-09-23)	Completed; results posted (outcome not retrieved here).
NCT07424404	KarXT; KarX-EC	Phase 3	Recruiting (2026-09-22)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT04630847	Fecal Microbiota Transplant	Phase 1	Not yet recruiting (2026-09-18)	Under investigation; results not yet available; effectiveness cannot be inferred.

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

1. **Wang et al. (2026)**, Meta-analysis: Exercise significantly improved overall, gross, and fine motor skills in children with ASD, with dose-response effects identified [1].
2. **Liu et al. (2026)**, Randomized controlled trial: A modified ketogenic diet + rehabilitation reduced ASD symptom severity (ABC/CARS scores) more than rehabilitation alone after 2 months [6].
3. **Seng et al. (2026)**, Randomized controlled trial: Caregiver-assisted group-based NDBIs showed early developmental improvements, though nonverbal gains were limited [5].
4. **Sikich et al. (2021)**, Randomized controlled trial (Phase II): Intranasal oxytocin did not significantly alter primary ABC-mSW scores vs. placebo in children/adolescents with ASD [8].

Conflicting evidence

None was retrieved.

What remains under investigation

- **Efficacy:** Head-to-head comparisons of exercise vs. ketogenic diet interventions for motor/social outcomes in ASD (no direct trials retrieved).
- **Effectiveness:** Long-term feasibility and scalability of caregiver-assisted group-based NDBIs in community settings (follow-up >2 years not reported).
- **Feasibility:** Optimal dosage and delivery formats for music therapy to improve social communication in ASD (protocol-only evidence retrieved).
- **Association:** Neurophysiological markers (e.g., sleep spindles) as predictors of treatment response (observational only; causality not established).
- **Effectiveness:** Water competency interventions for ASD-related drowning risks (protocol-stage trial; no efficacy data).

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages, not full texts; study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- Excluded retracted publication: Efficacy of oral folinic acid supplementation in children with autism spectrum disorder: a (PMID 39243316, 2024). Retracted work is not used as evidence.
- 1 retrieved record(s) were excluded because they were not about the question's subject.
- The PubMed / Europe PMC search timed out, so it may be incomplete; the absence of evidence cannot be concluded from it.

Sources

1. Wang T, Dong X, Jia Y, et al. Dose-response relationship between exercise interventions and motor skills in children with autism spectrum disorder: a systematic review and multilevel meta-analysis. *European journal of pediatrics*. 2026-09-18. [PMID 42758334](#). DOI 10.1007/s00431-026-07421-4.
2. Erçelebi H, Özbudak P, Menderes D, et al. Altered sleep spindle characteristics in children with autism spectrum Disorder: A potential neurophysiological marker. *Clinical neurophysiology : official journal of the International Federation of Clinical Neurophysiology*. 2026 Dec. [PMID 42753649](#). DOI 10.1016/j.clinph.2026.2112405.
3. Shi M, Hu J, Pei J, et al. Dual-target deep brain stimulation for severe pediatric autism: Sustained core symptom reduction in a 12-month case study. *Clinical neurophysiology : official journal of the International Federation of Clinical Neurophysiology*. 2026 Dec. [PMID 42759099](#). DOI 10.1016/j.clinph.2026.2112403.
4. Hitomi K, Katayama S, Nakada S, et al. Crohn's Disease Presenting as an Abdominal Wall Abscess in an Adolescent With Autism Spectrum Disorder. *Case reports in pediatrics*. 2026-09-22. [PMID 42781187](#). DOI 10.1155/crpe/8225541.
5. Seng GJ, Lin JY, Huang WL, et al. Feasibility and Preliminary Efficacy of a Caregiver-Assisted Group-Based Naturalistic Developmental Behavioral Intervention Program for Autistic Preschoolers: A Pilot Randomized Controlled Trial. *Autism : the international journal of research and practice*. 2026-07-23. [PMID 42494078](#). DOI 10.1177/13623613261462574.
6. Liu L, Zhao Q, Zhou J, et al. Evaluating the therapeutic efficacy of a modified ketogenic diet in children with autism spectrum disorder: a randomized controlled trial. *BMC pediatrics*. 2026-09-11. [PMID 42763398](#). DOI 10.1186/s12887-026-07708-3.
7. Hao D, Mansi H, Elkefi S, et al. Extended reality (XR) interventions in supporting children with autism: A systematic review. *International journal of medical informatics*. 2026 Dec. [PMID 42784985](#). DOI 10.1016/j.ijmedinf.2026.106719.
8. Sikich L, Kolevzon A, King BH, et al. Intranasal Oxytocin in Children and Adolescents with Autism Spectrum Disorder. *The New England journal of medicine*. 2021 Oct 14. [PMID 34644471](#). DOI 10.1056/NEJMoa2103583.
9. Randell E, Wright M, Milosevic S, et al. Sensory integration therapy for children with autism and sensory processing difficulties: the SenITA RCT. *Health technology assessment (Winchester, England)*. 2022 Jun. [PMID 35766242](#). DOI 10.3310/TQGE0020.

Other sources (regulators, guideline organisations, web pages, FDA labels)

- [Pharmacological treatments in autism spectrum disorder: a narrative review | Journal of Psychopathology](#)
- [FDA Takes Action to Make a Treatment Available for Autism Symptoms | FDA](#)
- [FDA Responds to Evidence of Possible Association Between Autism and Acetaminophen Use During Pregnancy | FDA](#)

- [FDA drug label: Aripiprazole \(label effective 2026-03-05\)](#)
 - [FDA drug label: Risperidone \(label effective 2026-06-25\)](#)
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Research support, not a substitute for clinical judgement. All citations, trial IDs, links and numbers in this report were checked against the retrieved sources.